

Git & Github

Basics :

- **Git** : Version Control System is a tools that helps to track changes in code.
- **Github** : Website that allows developers to store(in Repository) and manage their code using Git.
- **Process** : add -> Commit (on Github auto add and only need to commit)
- **~** : Root/Primary Folder
- **Pull Request** : It lets you tell others about changes you've pushed to a branch in a repo on Github.
- **Resolving merge conflicts** : An event that takes place when git is unable to automatically resolve differences in code between two commits.
- **Fork** : A fork is a new repo that shares code and visibility settings with the original "upstream" repo. (Rough copy)
- **Commands** :
 - cd (change dir)
 - cd .. (go to previous dir)
 - mkdir (make new dir)
 - ls (show readable file)
 - ls -a (show all file hidden also)
- **File stages** :
 - Untracked : new files that git doesn't yet track
 - Modified : changed
 - Staged : file is ready to be committed
 - Unmodified : unchanged
- **Basic Work flow** : Github repo - clone - changes - add - commit - push
Or Local repo - add - commit - push

Git Commands :

- **Configuring Git** :
 - `git config --global user.name "Our name"`
 - `git config --global user.email "Github email"`
 - `git config --list`
- **Clone (Cloning Repo on Local machine)** :
 - `git clone <-some link->`

- Status (Displays the state of the code) :
 - `git status`
- Add (new or changed files your working directory to the Git staging area)
 - `git add <-file name->`
 - `git add .` (add all files)
- Commit (It is a record of changes)
 - `git commit -m "changes related message"`
- Push (upload local repo content to remote repo(github))
 - `git push origin main` (origin - root remote repo, main - branch name)
- Init (create a new git repo) :
 - `git init`
 - `git remote add origin <-link->` (create remote repo with origin name)
 - `git remote -v` (to verify remote repo)
 - `git log` (check all commits)
 - `git log --one line` (in short form)
 - `git pull origin main` (fetch and download content from a remote repo and immediately update the local repo to match that content)
- git branches (different paths for same time commit)
 - `git branch` (to check branch)
 - `git branch -M main` (to rename branch)
 - `git checkout <-branch name->` (navigate on branch)
 - `git checkout -b <-new branch name->` (create new branch)
 - `git branch -d <-branch name->` (delete branch)
- Merge branches
 - Way 1 : `git diff <-branch name->` (difference)
 - `git merge <-branch name->` (merge 2 branches)
 - Way 2 : Create Pull Request
- Undoing changes
 - Case 1 : staged changes (only add)
 - `git reset <-file name->`

`git reset`

- Case 2 : committed changes (for one committed)

`git reset HEAD~1` (remove last commit)

- Case 3: committed changes (for many commits)

`git reset <-commit hash->` (only remove commit)

`git reset --hard <-commit hash->` (also remove code changes)